

PORTFOLIO PROJECT

CORPORACION FAVORITA DEMAND FORECAST

Predicting the future of retail. Minimizing risk. Maximizing profits.

16 Days

FORECAST HORIZON

28,512

VALIDATION ROWS

28,512

FUTURE ROWS

12.3M

PROJECTED TOTAL SALES

THE MARKET CHALLENGE

Short-term optimization for strategic replenishment.

- ⚡ **Exact Projection:** Forecast expected sales for each store and product family across the entire network.
- 🛡️ **Risk Mitigation:** Reduce the risk of stockouts, avoid excess inventory, and optimize holiday planning.
- 🧠 **Intelligent Context:** Leverage known future variables (calendar, metadata, promotions, and oil prices).
- 👁️ **Total Transparency:** Maintain an interpretable model to facilitate business review and portfolio storytelling.

600 × 400

OUR TECHNOLOGY: THE EDGE

LIGHTGBM RMSLE

0.4167

↗ Beats the baseline system

BASLINE RMSLE

0.5216

Traditional Operating System

FINAL MODEL COMPARISON BY WAPE (PERCENTAGE ERROR)



| ABSOLUTE PRECISION

LIGHTGBM RMSLE

0.0574

PROPHET RMSLE

0.1070

TOTAL FORECAST BIAS

-0.08%



OPPORTUNITY MAPPING

Key action zones to maximize profitability and monitor forecast error (WAPE):

 CRITICAL STORES (TOP RISK)

 KEY PRODUCT FAMILIES

Store 18

BOOKS

Store 25

HOME APPLIANCES

Store 32

BABY CARE

Store 38

SCHOOL AND OFFICE SUPPLIES

Store 42

HARDWARE

| A "TURNKEY" SYSTEM



API ARCHITECTURE

FastAPI exposes liveness, readiness, metadata, and batch prediction endpoints. Integrated into a Dockerfile for immediate deployment.



TOTAL SECURITY

Automated tests validate availability, model contract compatibility, and proper input error handling 24/7.



CI/CD PIPELINE

Continuous Integration that installs dependencies, compiles source code, runs tests, and builds the production Docker image on autopilot.

| THE ROADMAP TO SUCCESS

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STRATEGIC NEXT STEPS

-  Deploy the interactive **Power BI** report from the generated CSV layer, democratizing data.
-  Automate feature generation for raw queries (date, store, family), enabling non-technical users.
-  Register the final LightGBM model in the global **MLflow** registry and configure automated promotion gates.
-  Intensive post-launch performance monitoring in high-risk zones and peak seasons.

I IMAGE SOURCES

600 × 400

https://png.pngtree.com/png-clipart/20250708/original/pngtree-shiny-golden-arrow-curves-upwards-indicating-growth-and-progress-on-a-png-image_21290679.png
Source: [pngtree.com](https://png.pngtree.com/)

600 × 400

https://plus.unsplash.com/premium_photo-1733259831683-ff0e800097e2?fm=jpg&q=60&w=3000&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxzZWFiY2h8MXx8bmVIZCUyMGZvciUyMHNwZWVkfGVufDB8fDB8fHww
Source: unsplash.com

600 × 400

https://png.pngtree.com/thumb_back/fh260/background/20260112/pngtree-complex-network-structure-with-interconnected-nodes-and-lines-in-futuristic-3d-image_21066231.webp
Source: [pngtree.com](https://png.pngtree.com/)